Multivariate Specification Shifts in Linear TSCS Models

Joint, sequential, and leave-one-control-out decompositions

Technical derivation for the IVB paper

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Contents

1 Purpose and scope

This note extends the scalar specification-shift identity to a vector of candidate controls. It distinguishes three objects that answer different questions:

1. the **joint shift**, which compares a model containing none of the candidate controls with a model containing all of them;
2. **sequential increments**, which allocate that joint shift along one specified order of inclusion; and
3. **leave-one-control-out increments**, which compare the full model with a model that removes one control while retaining every other candidate control.

All three are contrasts between nested linear projections. They remain descriptive unless a causal target, a within-period ordering, a time-indexed DAG, and identification assumptions warrant a causal interpretation. In particular, neither the vector product nor any allocation of it automatically equals bias relative to the contemporaneous treatment effect (CET).

The formulas apply directly to linear TSCS specifications. Unit and time indicators, lagged outcomes, lagged treatments, and predetermined state variables can enter the common regressor vector W. Their presence does not change the algebra, but the algebra does not establish that the dynamics are correctly specified, that the within estimator is free of finite-T bias, or that any control is causally admissible.

2 Setup and notation

Let Y and D be scalar random variables, let W be a $k \times 1$ vector of common regressors that includes an intercept, and let

$$Z = (Z_1, \dots, Z_q)'$$

be a $q \times 1$ vector of candidate controls. Assume finite second moments and full rank for every projection used below. Define the short population linear projection by

$$(\beta_\emptyset, \gamma_\emptyset) = \arg\min_{b,g} \mathbb{E}\left[(Y - bD - W'g)^2\right], \tag{1}$$

and the joint long projection by

$$(\beta_{\mathcal{Z}}, \theta, \gamma_{\mathcal{Z}}) = \arg\min_{b,t,g} \mathbb{E}\left[(Y - bD - t'Z - W'g)^2\right], \tag{2}$$

where $\mathcal{Z} = \{1, \dots, q\}$ and θ is $q \times 1$. The joint population shift is

$$\Delta_{\mathcal{Z}} := \beta_{\mathcal{Z}} - \beta_\emptyset. \tag{3}$$

For any scalar or vector random variable A, write $\widetilde{A}_W = A - P_W A$ for the residual from its population linear projection on W. Define the vector auxiliary projection

$$Z = \pi D + KW + r, \qquad \mathbb{E}[Dr'] = 0, \qquad \mathbb{E}[Wr'] = 0, \tag{4}$$

where π is $q \times 1$ and K is $q \times k$. Component π_j is the coefficient on D in the auxiliary projection of Z_j on D and the common regressors W. By partial regression,

$$\pi = \frac{\mathbb{E}[\widetilde{Z}_W \widetilde{D}_W]}{\mathbb{E}[\widetilde{D}_W^2]}. \tag{5}$$

3 Joint vector identity

Proposition 1 (Joint multivariate specification shift)**.** *Under the stated moment and rank conditions,*

$$\boxed{\Delta_{\mathcal{Z}} = \beta_{\mathcal{Z}} - \beta_\emptyset = -\theta'\pi}. \tag{6}$$

This is an identity between coefficients of nested population linear projections.

Proof. Residualize Y, D, and Z with respect to W. The normal equation for D in the joint long projection is

$$\mathbb{E}[\widetilde{D}_W \widetilde{Y}_W] = \beta_{\mathcal{Z}} \mathbb{E}[\widetilde{D}_W^2] + \mathbb{E}[\widetilde{D}_W \widetilde{Z}_W'] \theta.$$

The short projection and Equation (5) give

$$\beta_{\emptyset} = \frac{\mathbb{E}[\widetilde{D}_W \widetilde{Y}_W]}{\mathbb{E}[\widetilde{D}_W^2]}, \qquad \mathbb{E}[\widetilde{D}_W \widetilde{Z}_W'] = \mathbb{E}[\widetilde{D}_W^2] \pi'.$$

Substitution in the long-model normal equation yields

$$\beta_{\emptyset} = \beta_{\mathcal{Z}} + \pi' \theta.$$

Because $\pi'\theta = \theta'\pi$ is scalar, rearrangement proves Equation (6). □

3.1 Projection-substitution verification

The sign can be checked independently. Write the joint long projection as

$$Y = \beta_{\mathcal{Z}} D + \theta' Z + W' \gamma_{\mathcal{Z}} + u_{\mathcal{Z}},$$

where $u_{\mathcal{Z}}$ is orthogonal to D, Z, and W. Substitute Equation (4):

$$Y = (\beta_{\mathcal{Z}} + \theta'\pi) D + W'(\gamma_{\mathcal{Z}} + K'\theta) + (u_{\mathcal{Z}} + \theta' r).$$

The composite residual is orthogonal to D and W. Uniqueness of the short projection therefore implies $\beta_{\emptyset} = \beta_{\mathcal{Z}} + \theta'\pi$, again giving the negative sign in the joint shift.

4 Exact sample identity

Let y and d be $n \times 1$ vectors, Z an $n \times q$ matrix, and W the common $n \times k$ design matrix. Let

$$M_W = I - W(W'W)^{-1}W'.$$

Assume W and the augmented design have full column rank and $d'M_W d > 0$. If $\hat{\theta}$ is the coefficient vector on Z in the joint long regression and

$$\hat{\pi} = \frac{Z' M_W d}{d' M_W d},$$

then the sample normal equation for d gives

$$\boxed{\widehat{\Delta}_{\mathcal{Z}} = \hat{\beta}_{\mathcal{Z}} - \hat{\beta}_{\emptyset} = -\hat{\theta}'\hat{\pi}}. \tag{7}$$

This equality is exact for OLS point estimates. It requires the same observations, weights, and common regressors in the short, joint long, and auxiliary regressions. Standard-error choices do not change it. With missing candidate controls, the short model must be re-estimated on the complete sample used by the joint model.

5 Sequential inclusion

Let $\sigma = (\sigma_1, \dots, \sigma_q)$ be a permutation of $\mathcal{Z}$. Define the nested sets

$$S_0 = \emptyset, \qquad S_k = \{\sigma_1, \dots, \sigma_k\}, \qquad k = 1, \dots, q,$$

and let β_{S_k} be the coefficient on D in the projection containing W and the controls indexed by S_k. The sequential increment assigned to the control entering at step k is

$$c_{\sigma_k|S_{k-1}} := \beta_{S_k} - \beta_{S_{k-1}}. \tag{8}$$

Let $\theta_{\sigma_k|S_{k-1}}$ be the coefficient on Z_{σ_k} in the outcome projection containing D, W, $Z_{S_{k-1}}$, and Z_{σ_k}. Let $\pi_{\sigma_k|S_{k-1}}$ be the coefficient on D in the auxiliary projection of Z_{σ_k} on D, W, and $Z_{S_{k-1}}$.

Proposition 2 (Path-specific sequential decomposition)**.** *For every order σ,*

$$c_{\sigma_k|S_{k-1}} = -\theta_{\sigma_k|S_{k-1}}\pi_{\sigma_k|S_{k-1}}, \tag{9}$$

and

$$\sum_{k=1}^{q} c_{\sigma_k|S_{k-1}} = \Delta_{\mathcal{Z}}. \tag{10}$$

The total is invariant to σ, but the individual increments generally are not.

Proof. At step k, absorb $Z_{S_{k-1}}$ into the common regressor vector. The scalar FWL identity applied to the addition of Z_{σ_k} gives Equation (9). Summing the direct coefficient differences gives

$$\sum_{k=1}^{q} (\beta_{S_k} - \beta_{S_{k-1}}) = \beta_{S_q} - \beta_{S_0} = \beta_{\mathcal{Z}} - \beta_{\emptyset},$$

because adjacent terms cancel. This proves telescoping. The conditional outcome slope and the conditional treatment–control association in Equation (9) both depend on S_{k-1}. With correlated controls, changing the order changes the variation assigned at each step, so the individual increments need not be invariant. □

For two controls, the two valid paths make the source of order dependence transparent:

$$c_{1|\emptyset} + c_{2|\{1\}} = \Delta_{\{1,2\}} = c_{2|\emptyset} + c_{1|\{2\}}. \tag{11}$$

Equation (11) constrains the sums, not the terms. Unless special covariance restrictions make the conditional and unconditional increments equal, $c_{1|\emptyset} \neq c_{1|\{2\}}$ and $c_{2|\emptyset} \neq c_{2|\{1\}}$.

6 Leave-one-control-out increments

For each $j \in \mathcal{Z}$, define

$$\ell_j := \beta_{\mathcal{Z}} - \beta_{\mathcal{Z}\setminus\{j\}}. \tag{12}$$

This is the increment from adding Z_j **last**, conditional on all other candidate controls. The scalar conditional identity gives

$$\ell_j = -\theta_{j|-j}\pi_{j|-j}, \tag{13}$$

where $\theta_{j|-j}$ is the coefficient on Z_j in the full outcome projection and $\pi_{j|-j}$ is the coefficient on D in the auxiliary projection of Z_j on D, W, and Z_{-j}.

Corollary 1 (Leave-one-out increments are not an additive decomposition)**.** *In general,*

$$\sum_{j=1}^{q} \ell_j \neq \Delta_{\mathcal{Z}}. \tag{14}$$

Proof. Each ℓ_j is a terminal edge from a different path through the lattice of nested models. These edges do not form a single path from $\emptyset$ to $\mathcal{Z}$, so their endpoints do not cancel. For $q = 2$, Equations (11) and (12) imply

$$(\ell_1 + \ell_2) - \Delta_{\{1,2\}} = (c_{1|\{2\}} + c_{2|\{1\}}) - (c_{1|\emptyset} + c_{2|\{1\}}) = c_{1|\{2\}} - c_{1|\emptyset}.$$

There is no projection identity forcing the final difference to be zero. Equality can occur under special covariance restrictions, but it is not guaranteed. With correlated controls, the terminal comparisons can allocate shared predictive variation more than once or omit variation that appears only along another path. □

Thus the three diagnostics should not be interchanged. The joint shift measures total movement across the two endpoint models. A sequential decomposition allocates that movement along one chosen path and therefore sums by construction. A leave-one-out increment measures the conditional sensitivity to deleting one control from the full model and is not an allocation of the joint total.

7 Deterministic numerical verification

The accompanying R script constructs 240 deterministic observations from trigonometric sequences. It is not a Monte Carlo experiment and does not execute any simulation used by the paper. The design contains two common regressors, one focal treatment, and three correlated candidate controls. Every projection uses the same observations and unweighted OLS. The test tolerance is 10^{-12}, which is stricter than 10^{-10}.

Table 1: Correlation matrix in the deterministic numerical design.

	D	Z1	Z2	Z3
D	1.0000	0.9013	0.5152	0.9086
Z1	0.9013	1.0000	0.7324	0.7360
Z2	0.5152	0.7324	1.0000	0.2761
Z3	0.9086	0.7360	0.2761	1.0000

The correlation matrix confirms that the candidate controls share substantial variation with one another and with the focal treatment. This makes the allocation problem nontrivial while retaining full column rank.

Table 2: Joint multivariate specification-shift identity.

quantity	value
$\hat{\beta}_{\emptyset}$	2.466118485285
$\hat{\beta}_{\mathcal{Z}}$	1.260540966936
$\widehat{\Delta}_{\mathcal{Z}}$	-1.205577518349
$-\hat{\tilde{\theta}}'\hat{\pi}$	-1.205577518349
Absolute identity error	4.441e-16

Table 3: Sequential contributions telescope for every order of three controls.

Order	Sequential sum	Joint shift	Absolute error
Z1 -> Z2 -> Z3	-1.205577	-1.205577	0.00e+00
Z1 -> Z3 -> Z2	-1.205577	-1.205577	1.55e-15
Z2 -> Z1 -> Z3	-1.205577	-1.205577	0.00e+00
Z2 -> Z3 -> Z1	-1.205577	-1.205577	8.88e-16
Z3 -> Z1 -> Z2	-1.205577	-1.205577	1.33e-15
Z3 -> Z2 -> Z1	-1.205577	-1.205577	1.78e-15

Table 4: Order dependence of control-specific sequential increments.

Control	Minimum increment	Maximum increment	Range across orders
Z1	-0.8487615	-0.24825506	0.6005064
Z2	-0.3787244	0.01218349	0.3909079
Z3	-0.8904769	-0.36899951	0.5214774

Every order reaches the same endpoint shift, while the increment attributed to each control changes with the preceding set. This is the empirical content of telescoping with order-dependent allocations.

Table 5: Leave-one-control-out increments and their conditional product identities.

Control	Direct leave-one-out increment	Conditional product	Absolute identity error
Z1	-0.32728413	-0.32728413	7.22e-16
Z2	-0.08698809	-0.08698809	1.33e-15
Z3	-0.36899951	-0.36899951	1.11e-16

In this deterministic design, the leave-one-out increments sum to -0.7832717315, whereas the joint shift is -1.205577518. Their difference is 0.4223057869. Each leave-one-out comparison satisfies its own conditional scalar identity, but the collection of terminal comparisons does not telescope.

Table 6: Deterministic verification summary.

Check	Diagnostic	Criterion	Result
Joint vector identity	4.441e-16	error < 1e-12	PASS
Conditional scalar identities	1.332e-15	error < 1e-12	PASS
Sequential telescoping, all orders	1.776e-15	error < 1e-12	PASS
Controls are correlated	2.761e-01	minimum absolute pairwise correlation > 0.20	PASS
Sequential allocations vary by order	3.909e-01	minimum range across orders > 1e-4	PASS
Leave-one-out sum differs from joint shift	4.223e-01	absolute gap > 1e-4	PASS

8 Interpretation and boundaries

The joint product $-\theta'\pi$ is the multivariate analogue of the scalar diagnostic. Its terms combine the conditional association between each candidate control and the outcome with the association between that control and treatment after partialling out W. When controls are correlated, however, the coordinate products should not be read as a unique allocation of the joint shift: θ is estimated jointly, while each component of π comes from an auxiliary projection conditional only on W. The scalar sum is invariant to a nonsingular reparameterization of the span of Z, but its coordinate-level representation is basis dependent.

Sequential increments answer a path-specific question. They are useful when the order has a substantive justification, such as a prespecified sequence from pretreatment history to increasingly contemporaneous measurements. Without that justification, an order-specific allocation is not unique. Leave-one-out increments answer a different robustness question: how much the treatment coefficient changes when one control is removed from an otherwise full specification. They are conditional sensitivities and should not be normalized as though they were shares of the joint shift.

None of these statements turns a coefficient movement into causal bias. If the endpoint models differ in their discrepancies from β_{CET}, then the joint shift is the difference between those discrepancies. Determining whether the movement is deconfounding, collider bias, over-control, or a mixture requires the causal clock, the role of every candidate control, and identification conditions outside the FWL algebra.

9 Reproducibility record

The numerical verification is implemented in `scripts/test_multivariate_specification_shift.R`. It uses base R only, contains no random-number generation, and stops with an error unless all joint, conditional, and telescoping identities satisfy the stated tolerance. The script also requires nontrivial control correlation, nonzero order dependence, and a nonzero gap between the leave-one-out sum and the joint shift. No file in `simulations/` or `replication/` is read or modified.